Restoration-Guided State-Space Adaptation for Degradation-Robust RGB Object Tracking

Abdel Rahman Idrais, *Student Member, IEEE*, Alireza Esmaeilzehi, *Member, IEEE*, M. Omair Ahmad, *Fellow, IEEE*, and M. N. S. Swamy, *Fellow, IEEE*

***Abstract*—Visual object trackers can degrade under image-quality corruptions such as blur, low resolution, and sensor noise. This paper studies a conservative restoration-guided state-space adaptation of OSTrack for RGB template-search tracking [1]. The final Paper Freeze V1 method inserts a Restoration-Guided State-Space Block (RG-SSB) into OSTrack, keeps the backbone frozen, and trains only `rgssb.*` and `box_head.*` using degradation-aware LaSOT training with global clean–degraded feature consistency at weight $\lambda = 0.02$ [2]. Response consistency, target-region feature consistency, and target-versus-distractor margin loss are not retained in the final method. On selected benchmark subsets, the method improves average AUC on broader OTB by $+0.033144$ and expanded UAV123 by $+0.017405$, while corrected expanded NFS is slightly negative at -0.009638. The cross-benchmark aggregate is near-neutral positive at $+0.001478$. These results support a cautious conclusion: restoration-guided feature adaptation shows robustness potential, but it does not establish state-of-the-art or universal degradation robustness.**

***Index Terms*—Visual object tracking, degradation robustness, state-space models, Mamba, feature consistency, RGB tracking.**

I. Introduction

Single-object tracking is a fundamental component of video analysis systems that must continuously localize a target after it has been specified in an initial frame. In practical deployments, the tracker is expected to operate under camera motion, target deformation, background clutter, partial occlusion, and changes in imaging quality. Benchmarks such as OTB, UAV123, LaSOT, and NFS have therefore become important tools for evaluating localization behavior under diverse sequence conditions [2]–[5]. For RGB template-search tracking, however, the problem is not only to model target appearance over time, but also to preserve discriminative target information when the search image is degraded by blur, reduced resolution, noise, or other acquisition artifacts.

Abdel Rahman Idrais, M. Omair Ahmad, and M. N. S. Swamy are with the Department of Electrical and Computer Engineering, Concordia University, Montreal, QC H3G 1M8, Canada.

Alireza Esmaeilzehi is with The Edward S. Rogers Sr. Department of Electrical and Computer Engineering, University of Toronto, Toronto, ON M5S 3G8, Canada.

Corresponding author: Abdel Rahman Idrais. Email structure requires confirmation; local author-information text extraction renders the address as a idrais@encs.concordia.ca.

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Recent one-stream trackers have shown that joint template-search feature learning can provide an effective and compact tracking formulation. OSTrack is a representative example, since it learns relations between the template and search regions within a unified feature stream and is used as the baseline architecture in this work [1]. Such trackers can be effective under clean or moderately challenging observations, but degraded search frames may alter the feature statistics on which template-search matching and box regression depend. This vulnerability is especially relevant when the tracker is trained and evaluated under clean and synthetically degraded branches, because the model must localize the same target even when the degraded search feature is less consistent with its clean counterpart.

Several lines of work are relevant to this robustness problem. Low-light and nighttime tracking studies show that degraded illumination is already an active tracking problem [6], [7]. Image restoration methods, including recent state-space restoration models, also aim to recover useful visual structure from degraded observations [8], [9]. These directions are valuable, but they do not directly answer the narrower question considered here. A tracker can be made degradation-aware without using restoration-oriented state-space adaptation, and an image restoration model can improve image quality without being optimized for template-search localization.

State-space models and Mamba-based vision modules have recently attracted attention because they provide a mechanism for structured long-range feature modeling with favorable computational properties [10]–[12]. In visual tracking, reported Mamba-based designs use state-space modules for purposes such as long-term context modeling, multimodal feature interaction, nonlinear motion prediction, dynamic template generation, or nighttime template-search representation learning [7], [13]–[16]. In parallel, MambaIR and MambaIRv2 show that state-space modeling can be adapted to restoration tasks through restoration-oriented feature processing [8], [9]. The unresolved gap addressed in this paper is therefore specific: whether a restoration-guided state-space block can be inserted into an RGB template-search tracker to improve clean–degraded feature consistency, without replacing the tracking framework or claiming a general solution to degradation-robust tracking.

To examine this question under controlled conditions, this paper introduces a Restoration-Guided State-Space Block (RG-SSB) into OSTrack. The module is placed after the frozen backbone search features and before the box head, and the final training configuration updates only `rgssb.*`

and `box_head.*`. The method uses global clean–degraded feature consistency with weight $\lambda = 0.02$, while response consistency, target-region feature consistency, target-versus-distractor margin loss, degradation tokens, memory update, response fusion, and template-guided scan are not retained in the final method. This design deliberately separates restoration-guided feature adaptation from image reconstruction, from ordinary preprocessing, and from broader changes to the tracking architecture.

The main contributions of this work are as follows:

- A conservative OSTrack+RG-SSB tracking adaptation is introduced, in which restoration-guided state-space processing is integrated between the frozen backbone features and the box head. This isolates the effect of the proposed block from full-backbone retraining and broad tracker redesign.
- A degradation-aware training objective is evaluated using global clean–degraded feature consistency at $\lambda = 0.02$. The final configuration is supported by the method-selection evidence, while response consistency, target-region feature consistency, and target-versus-distractor margin loss are reported as rejected alternatives rather than components of the proposed method.
- The method is evaluated under clean, motion-blur, low-resolution, and Gaussian-noise conditions on selected OTB, UAV123, and corrected NFS subsets [3]–[5]. The evidence supports positive average AUC changes on the selected OTB and UAV123 subsets, but corrected NFS remains mixed/slightly negative, so the claims are limited to a cautious proof of concept.
- The experimental record includes corrected NFS annotation provenance, same-GPU efficiency and parameter comparisons, and explicit claim boundaries. These checks prevent the corrected results from being combined with superseded NFS metrics and avoid state-of-the-art or universal robustness claims.

The remainder of this paper is organized as follows. Section II reviews related work and positions the proposed method. Section III describes the RG-SSB formulation and training objective. Sections IV and V present the experimental protocol and results, and Sections VI and VII discuss limitations and conclude the paper.

II. Related Work

A. Transformer and One-Stream Visual Tracking

Recent RGB trackers commonly formulate single-object tracking as template-search interaction. Early transformer trackers used attention to connect the initial target template with the current search region and to improve discriminative relation modeling [17], [18]. Subsequent one-stream trackers further reduced the separation between feature extraction and relation modeling. MixFormer uses mixed attention for simultaneous feature extraction and target-information integration, while OSTrack unifies feature learning and relation modeling by passing template and search tokens through a shared stream [1], [19].

Other recent trackers explore complementary formulations. SeqTrack treats tracking as sequence generation, ROMTrack models inherent and hybrid template features, and HIPTrack uses historical prompts [20]–[22].

ODTrack propagates dense temporal tokens online, LoRAT adapts large vision transformers with low-rank fine tuning, and AQATrack uses autoregressive spatio-temporal queries [23]–[25].

These methods show that modern tracking performance often depends on how target information is represented, propagated, and fused with the search region. They also provide the architectural context for this paper. The present work does not propose a new transformer tracker, a new generative tracking head, or a new large-backbone adaptation strategy. Instead, it keeps the OSTrack template-search formulation and studies whether the already extracted search representation can be adapted under degraded observations.

B. Temporal Modeling and Target Adaptation

Temporal information is a major source of robustness in visual tracking because target appearance, background clutter, and localization uncertainty change over time. STARK introduced spatio-temporal transformer tracking, while HIPTrack and ODTrack show two recent directions for using tracking history through historical prompts or dense temporal token propagation [18], [22], [23]. SeqTrack and AQATrack also connect tracking with autoregressive or spatio-temporal query mechanisms, replacing part of the traditional hand-designed tracking pipeline with sequence or query prediction [20], [25]. These designs are relevant because they improve target adaptation without requiring the present paper's restoration-guided state-space feature block.

The distinction is important for the proposed method. Temporal prompts, autoregressive queries, and online token propagation aim to exploit historical target information or simplify tracking prediction. They do not by themselves establish whether a degraded search feature can be made more consistent with its clean counterpart after search-token extraction. The final RG-SSB configuration therefore avoids memory update, response fusion, target-region consistency, and target-versus-distractor margin loss, because the retained evidence supports a narrower global clean–degraded feature-consistency setting rather than a broader temporal-adaptation design.

C. State-Space Models for Visual Representation

State-space sequence modeling has recently become an alternative mechanism for long-range dependency modeling. Mamba introduces selective state spaces with input-dependent parameters and hardware-aware computation for efficient sequence modeling [10]. Vision Mamba adapts bidirectional Mamba blocks to visual representation learning, and VMamba develops visual state-space blocks with two-dimensional selective scanning for vision backbones [11], [12]. These works are not tracking methods, but they are important because they show how state-space models can be reorganized for visual tokens rather than only for one-dimensional language or signal sequences.

Restoration-oriented Mamba models adapt this family in a different direction. MambaIR addresses low-level restoration with residual state-space blocks, local enhancement, and channel attention, while MambaIRv2 develops attentive state-space restoration and semantic-guided neighboring to reduce limitations of causal scanning in image restoration [8], [9]. These papers support the technical motivation for using state-space processing under degraded visual evidence. However, their objectives are image restoration objectives, not template-search localization objectives. A tracker needs target-discriminative features and accurate box prediction, so an external restoration network or full image-reconstruction objective cannot be assumed to improve tracking.

D. Mamba-Based Visual Tracking

Mamba-based tracking has already been explored, so the present paper does not claim novelty from simply using Mamba in tracking. MambaLCT uses a Context Mamba module to aggregate long-term target variation cues over video frames [13]. Mamba-FETrack and Mamba-FETrack V2 use Mamba-style backbones and fusion mechanisms for RGB-event tracking, where event data provide complementary sensing under difficult imaging conditions [14], [26]. MambaEVT studies event-stream tracking with Vision Mamba and Memory Mamba for dynamic template generation [16]. MambaNUT applies a pure Mamba one-stream design with adaptive curriculum learning for nighttime UAV tracking [7].

These works demonstrate several uses of state-space modeling in tracking: temporal context aggregation, multimodal RGB-event fusion, event-only dynamic template generation, efficient backbone design, and nighttime template-search learning. They are therefore direct novelty boundaries for this paper. The proposed RG-SSB is not a complete Mamba backbone, not an RGB-event or event-only tracker, not a temporal-memory module, and not a nighttime-specific curriculum tracker. Its placement is more limited: it is inserted after search-token extraction in OSTrack to adapt degradation-sensitive search representations while retaining the frozen backbone and training only `rgssb.*` and `box_head.*`.

E. Tracking Under Degraded Observations and Present-Work Positioning

Tracking under degraded observations has been studied through benchmark construction and adverse-condition tracking. LLOT explicitly targets low-light object tracking and reports a benchmark intended to expose performance gaps under low illumination [6]. MambaNUT further confirms that nighttime UAV tracking can be handled as a specific low-light/adverse-condition problem without using a separate restoration module [7]. Thus, the manuscript does not claim that low-light tracking or nighttime UAV tracking is new.

The present contribution is instead a controlled test of a narrower intersection. It differs from replacing OSTrack with a complete Mamba tracker, from multimodal Mamba trackers that rely on event data, from temporal-memory Mamba trackers, from nighttime-specific Mamba trackers, from image restoration used as external preprocessing, and from full restoration networks optimized for image quality. The proposed RG-SSB adapts search features inside the tracker after search-token extraction and uses global clean–degraded feature consistency with weight $\lambda = 0.02$. This positioning supports only a limited claim: restoration-guided state-space adaptation can be evaluated as a feature-level modification to OSTrack under selected RGB degradations. It does not establish state-of-the-art tracking performance, universal degradation robustness, or uniform improvement across all benchmarks.

III. Proposed Method

A. Overview

The final method uses OSTrack as the base tracker. Given a template image and a search image, OSTrack extracts transformer features and predicts a response map and bounding box. Paper Freeze V1 adds a Restoration-Guided State-Space Block (RG-SSB) after the backbone search features and before the tracking head. The module is designed to adjust tracking features under degradation rather than reconstruct RGB images.

The frozen method is:

- base tracker: OSTrack;
- added module: RG-SSB;
- frozen parameters: backbone;
- trainable parameters: `rgssb.*` and `box_head.*`;
- training data: degradation-aware LaSOT [2];
- feature consistency: global clean–degraded feature consistency;
- feature-consistency weight: $\lambda = 0.02$;
- disabled components: response consistency, target-region feature consistency, target-versus-distractor margin loss, degradation tokens, memory update, response fusion, and template-guided scan.

B. Training Objective

The training objective combines the degraded-branch tracking loss with global clean–degraded feature consistency:

$$\mathcal{L} = \mathcal{L}_{\text{track}}(X_d) + 0.02\,\mathcal{L}_{\text{feat}}(F_c, F_d), \tag{1}$$

where X_d is the degraded search branch, and F_c and F_d are clean and degraded post-RG-SSB search features. The clean feature branch is detached in the consistency term. The final method intentionally keeps the response-consistency, target-region feature-consistency, and target-versus-distractor margin (TDM) objectives disabled.

IV. Experimental Setup

A. Baseline and Final Configuration

The baseline is original OSTrack config `vitb_256_mae_ce_32x4_ep300`. The final method uses `vitb_256_mae_ce_32x4_ep300_rgssb_head_train_lasot_degraded_hpc_featcons_lam002`. All reported changes are computed as RG-SSB minus the OSTrack baseline for the same sequence, degradation, severity, and seed.

B. Benchmarks and Degradation Conditions

The evaluated subsets are:

- broader OTB: 13 sequences and 52 sequence-condition pairs [3];
- expanded UAV123: 16 sequences and 64 pairs [4];
- corrected expanded NFS: 32 sequences and 128 pairs [5].

Each sequence is evaluated under four conditions: clean, motion blur at medium severity, low resolution at medium severity, and Gaussian noise at medium severity. Degraded conditions use seed 42.

C. Metrics

The paper reports Success AUC, Precision@20, mean center error, model parameters, controlled FPS, per-frame latency, and peak allocated GPU memory. FLOPs/MACs are unavailable and are not reported as measured.

D. Corrected NFS Provenance

NFS required a correction before current results could be used. Raw NFS annotations were confirmed to contain XYXY coordinates, while the previous invalid path treated aligned rows as XYWH. The invalid NFS rows were archived, normalized aligned XYWH annotations were prepared, and NFS tracking was rerun. The current NFS metrics and failure analysis use only corrected normalized aligned XYWH annotations. Earlier NFS metrics and qualitative overlays are superseded.

V. Results

A. Main Benchmark Results

Table I summarizes the benchmark-level results, and Fig. 1 visualizes the same average AUC changes. The strongest result is on OTB, where RG-SSB improves average AUC by $+0.033144$ across 52 sequence-condition pairs. UAV123 is also positive, with $+0.017405$ average AUC over 64 pairs. Corrected NFS is the main weakness: the method decreases average AUC by -0.009638 over 128 pairs. The combined cross-benchmark aggregate is $+0.001478$, which is best described as near-neutral positive rather than consistent superiority.

B. Condition-Level Results

Table II and Fig. 2 summarize condition-level behavior. The method is most consistently positive under motion blur. Low resolution remains the clearest condition-level weakness, with -0.008275 average AUC and increased center error.

C. Ablation Studies

Table III reports the final method-selection ablations, and Fig. 3 visualizes the decisions. Global feature consistency with $\lambda = 0.02$ is retained. Response consistency, target-region feature consistency, and TDM are rejected. TDM is important negative evidence: it improves several local sequence-condition pairs, but its Gaussian-noise regression dominates the aggregate gate result. It is therefore not part of the frozen method.

The final TDM gate decision is `REJECT_AND_ROLL_BACK_TO_PAPER_FREEZE_V1`.

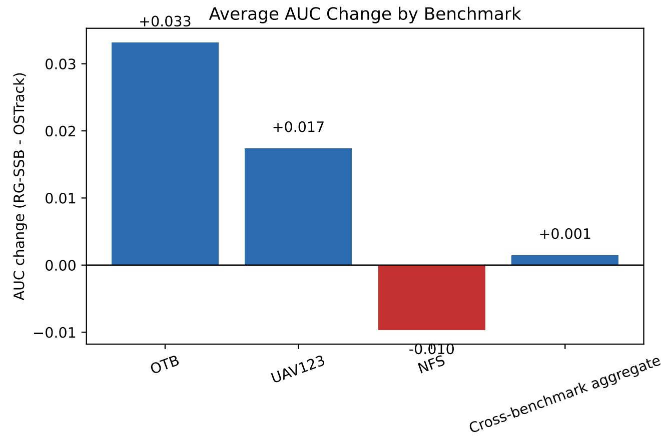

Fig. 1. Benchmark-level AUC change for Paper Freeze V1. Positive values favor RG-SSB. The evaluated scope is selected OTB, expanded UAV123, corrected expanded NFS, and the cross-benchmark aggregate.

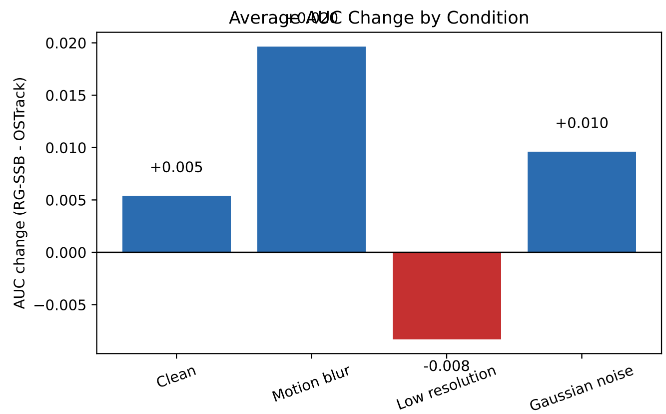

Fig. 2. Condition-level AUC change across clean, motion blur, low resolution, and Gaussian noise. The low-resolution condition remains mixed/negative.

D. Efficiency Analysis

Table IV and Fig. 4 report the same-GPU controlled efficiency comparison. The measurement uses a Tesla V100-PCIE-32GB, the same Car1 clean sequence, 50 warm-up frames, 3 repetitions, synchronized CUDA timing, and separate initialization/per-frame tracking measurements. Historical timing files are not mixed with this controlled comparison.

The final method adds 2,079,936 parameters over the baseline. FLOPs/MACs are unavailable and are not reported as measured.

E. Corrected NFS Failure Analysis

Figure 5 summarizes terminal outcomes for 10 corrected NFS failure cases selected from the largest corrected drops. Persistent RG-SSB-specific target loss appears in 5 cases, sudden center jump in 3 cases, shared tracker failure in 1 case, and ambiguous behavior in 1 case. This is a qualitative failure-focused analysis, not a benchmark-wide frequency estimate.

VI. Discussion and Limitations

The evidence supports a narrow, defensible claim: restoration-guided state-space adaptation can improve selected

TABLE I

PAPER FREEZE V1 BENCHMARK SUMMARY. POSITIVE AUC AND PRECISION@20 CHANGES FAVOR RG-SSB; NEGATIVE CENTER-ERROR CHANGES INDICATE LOWER ERROR.

Benchmark	Sequences	Pairs	Baseline AUC	RG-SSB AUC	AUC Change	Precision Change	Center Error Change	Conclusion
OTB	13	52	0.646964	0.680108	+0.033144	+0.050250	-4.563202	Improves clearly on average
UAV123	16	64	0.660183	0.677587	+0.017405	+0.029983	-3.500185	Improves slightly / near neutral
NFS	32	128	0.681149	0.671511	-0.009638	-0.006937	+3.705562	Mixed / negative average
Cross-benchmark aggregate	53	212	0.689269	0.690748	+0.001478	+0.007843	+1.161416	Near-neutral positive overall

TABLE II

CONDITION-LEVEL ROBUSTNESS SUMMARY ACROSS SELECTED OTB, EXPANDED UAV123, AND CORRECTED EXPANDED NFS SUBSETS.

Condition	Pairs	Baseline AUC	RG-SSB AUC	AUC Change	Precision Change	Center Error Change	Conclusion
Clean	61	0.685280	0.690648	+0.005368	+0.015220	+2.558046	Improves on average
Motion blur	61	0.643687	0.663299	+0.019612	+0.029967	-7.502857	Improves on average
Low resolution	61	0.672450	0.664175	-0.008275	-0.003989	+6.769353	Mixed / negative
Gaussian noise	61	0.672042	0.681627	+0.009585	+0.018540	-0.342091	Improves on average

TABLE III

ABLATION SUMMARY FOR PAPER FREEZE V1. LOCAL ABLATIONS GUIDE METHOD SELECTION AND ARE NOT FULL BENCHMARK CLAIMS.

Ablation	Scope	Primary Comparison	Average AUC Change	Decision	Evidence
Global feature consistency $\lambda = 0.02$	3 OTB sequences $\times$ 4 conditions	vs. OSTrack baseline	+0.023480	Kept	Lambda sweep CSV
Response consistency	3 OTB sequences $\times$ 4 conditions	vs. feature-only $\lambda = 0.02$	-0.019360	Rejected	Response-consistency CSV
Target-region feature consistency	3 OTB sequences $\times$ 4 conditions	vs. global feature consistency	-0.023555	Rejected	Target-vs-global CSV
Target-vs-distractor margin	3 OTB sequences $\times$ 4 conditions	vs. global feature consistency	-0.014549	Rejected by gate	TDM gate CSV

TABLE IV

CONTROLLED SAME-GPU EFFICIENCY COMPARISON ON TESLA V100-PCIE-32GB. FLOPS/MACS WERE UNAVAILABLE AND ARE NOT REPORTED AS MEASURED.

Model	Params	Trainable Params	Frozen Params	Init. ms	Latency ms	FPS	Peak Alloc. MB	Peak Reserved MB	FLOPs/MACs
OSTrack baseline	92,518,533	92,518,533	0	2.949	10.724	93.247	380.233	434.000	Unavailable
RG-SSB final	94,598,469	8,554,053	86,044,416	3.165	11.525	86.766	388.917	434.000	Unavailable

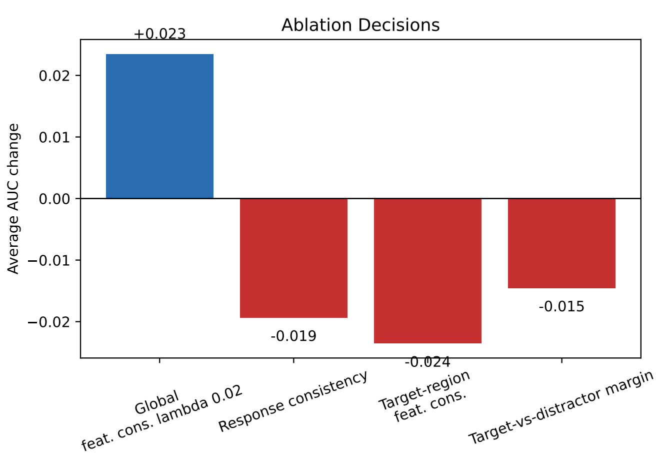

Fig. 3. Ablation decision summary. TDM is reported only as rejected negative evidence and is not part of Paper Freeze V1.

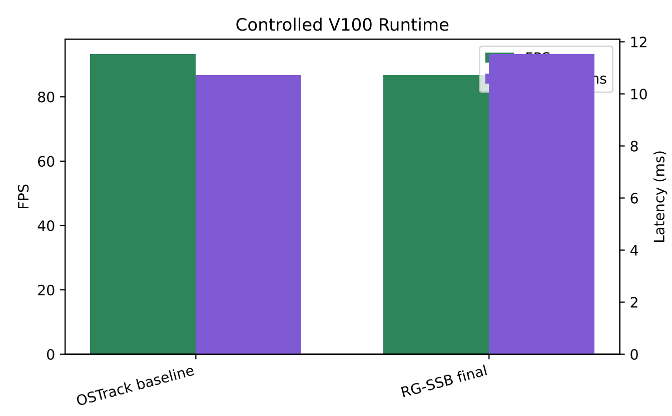

Fig. 4. Controlled V100 runtime comparison. The final method has a modest FPS reduction relative to the baseline and a small increase in peak allocated memory.

robustness outcomes in RGB template-search tracking, especially on the tested OTB subset. UAV123 is positive but mixed, and corrected NFS is slightly negative. The result is therefore promising but not uniformly improved.

The low-resolution weakness and corrected NFS failures are important limitations. They suggest that global feature consistency may not fully protect target identity under fast motion, high-frame-rate sampling behavior, or severe target/background ambiguity. The rejected TDM ablation further shows that adding a response-margin objective is not automatically beneficial; poorly balanced response constraints can introduce large condition-specific regressions.

This paper version has several boundaries. The benchmark coverage uses selected subsets, not complete benchmark suites. Degradations are synthetic, medium severity, and use one seed.

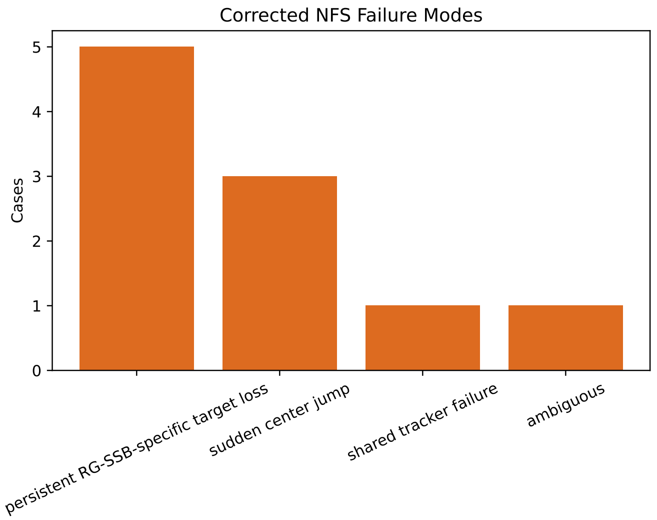

Fig. 5. Corrected NFS failure-mode distribution for 10 selected failure cases. Ground truth and diagnostics use corrected aligned XYWH annotations.

Training uses LaSOT only. The backbone remains frozen. FLOPs/MACs are unavailable. Corrected NFS required annotation normalization and rerunning NFS tracking; older NFS metrics and overlays are superseded. The evidence does not support state-of-the-art claims, universal degradation robustness, or uniform gains across all benchmarks.

VII. Conclusion

Paper Freeze V1 establishes a conservative fallback method: OSTrack with RG-SSB and box-head training under global clean–degraded feature consistency. The method is positive on OTB and UAV123, mixed on corrected NFS, and near-neutral positive in the mandatory cross-benchmark aggregate. This is sufficient for a cautious proof-of-concept paper position, but not for claims of broad superiority. Future work should expand benchmark coverage, test multiple seeds and severities, and investigate failure-specific refinements only after the current evidence is presented with appropriate limits.

Appendix A
Reproducibility Notes

The final method config is `vitb_256_mae_ce_32x4_ep300_rgssb_head_train_lasot_degraded_hpc_featcons_lam002`. The frozen method keeps the backbone frozen and trains only `rgssb.*` and `box_head.*`. NFS results must use the corrected normalized aligned XYWH annotation root. The invalid historical NFS rows are archived for provenance only and must not be used for current paper claims. FLOPs/MACs are unavailable and should not be reported as measured.

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